

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 2 EXAMINATION 2021-2022****CE4172 – Internet of Things: Tiny Machine Learning**

Apr/May 2022

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains 5 questions and comprises 2 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions do NOT carry equal marks.

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1. (a) State the main advantages of TinyML. (8 marks)
 - (b) What are memory bound and compute bound applications, and how are they relevant in the context of TinyML deployment? (15 marks)
 2. Briefly explain how we can identify whether a trained model is overfitting, and the steps that we can use to reduce overfitting of the model. (12 marks)
 3. (a) What are the main differences between TensorFlow and TensorFlow Lite deep learning frameworks? (10 marks)
 - (b) Briefly explain the typical steps used to deploy a trained model on a microcontroller in TinyML. (15 marks)

4. (a) Briefly explain the importance of optimizing the latency in TinyML applications.
(10 marks)
- (b) What are the techniques that can be used to reduce the latency?
(15 marks)
5. What is Cascade Design? Use an example to illustrate the concept of Cascade Design for TinyML deployment.
(15 marks)

END OF PAPER

CE4172 INTERNET OF THINGS: TINY MACHINE LEARNING

Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.